Mathematical and computational supplement to An exotic $S^2 \times S^2$ from a certificate-checked surface-bundle surgery manifold

John Clyde
VentiMath
verification@ventimath.org

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Abstract

This supplement preserves the implementation inventory, attribution checklist, machine-checked dependency chains, trust boundary, data pins, calibrations, and superseded exports behind the main paper. The article proves that $\pi_1(V_{\mathrm{aud}}) = 1$ and that the double Z_{aud} of V_{aud} along its intrinsic boundary involution is a closed symplectic four-manifold homeomorphic but not diffeomorphic to $S^2 \times S^2$. The comparison $V_{\mathrm{aud}} \cong V$ with Wuebben's member is not a theorem there and enters no proof: clauses D1–D14 below record what that attribution must establish. The existence chain for Theorem A$'$ carries no assumption; the earlier transfer of Wuebben's Theorems A, B, C under D1–D14 is retained, unchanged, as the attribution chain.

How to use this supplement

The main paper contains the mathematical certificate specification, the product-filled audit-model identification, the intrinsic topology of V_{aud}, a separate product-to-Lagrangian framing theorem, the intrinsic boundary involution σ_{aud}, and the proof that the double Z_{aud} is an exotic $S^2 \times S^2$. Section 1 preserves the full attribution ledger, and Section 2 below records the concrete certificate batches and replay architecture. Appendix A describes the assumption-free existence chain and preserves the historical transfer chain; Appendix B records provenance, trust, and data availability; Appendix C preserves calibrations and historical computations. The machine-readable ledger and every bound digest remain normative for exact inventories.

1 Open source-comparison checklist D1–D14

The main paper deliberately makes no transfer theorem about Wuebben's fixed member: its Theorem A$'$ is about Z_{aud}, and no clause below is a hypothesis of it. The full clause-by-clause attribution checklist follows; the geometric layer inventory, including filenames and checker sizes, remains in Section 3. D1–D11 are textual or diagrammatic identifications to be checked against the source, while D12–D14 are the genuine relative smooth and framing comparison problem.

Table S1: Detailed source-to-audit comparison dictionary. The final column, rather than the supporting evidence, is the conditional boundary.

	Target location	Audit interpretation	Mechanical or cited support	Residual assertion
D1	[1, Convention 2.1(C8), §7.1, Fig. 1]	ordered curves are the audit five-chain (a, b, c, d, e)	independent text extraction; ribbon intersection and filling checks; source-figure hash	textual transcription of labels and order
D2	[1, Lemma 7.1]	chain-reversing involution exchanges $a \leftrightarrow e$, $b \leftrightarrow d$, preserves c, and has fixed points p, O	exhaustive equivariant ribbon rotations; periodic-disk normal form; independent subdivision	diagrammatic identification of the marked complementary disks
D3	[1, §7.1, Fig. 1] and imported source figure	hidden, dashed, or paired segments continue displayed curves rather than adding crossings	vector-source and planar-incidence extraction; mutation controls	diagrammatic continuation convention
D4	[1]; Conv. 2.1 (C2,C5); §§3.1,7.2	$\psi_0 = T_a \circ T_b$ applies T_b first with the displayed right-twist normalization	based action in two subdivisions; reversed-order and reversed-sign controls	textual convention and realization by the chosen representatives
D5	[1, Convention 2.1(C4), §7.3]	cut square assigns φ_0 to α holonomy and ψ_0 to β holonomy	explicit graph-clutching map and inverse; seam checks	textual cut and orientation convention
D6	[1, §§7.2–7.3]	T_α is the c-over-α subbundle and T_β the e-over-β product subbundle, relative to full collars	relative-monodromy certificates; disjoint induced-torus and local-flatness checks	identification of target curve subbundles with the audit tori
D7	[1]; Conv. 2.1 (C1,C3,C6)	paths compose left to right; $A = [\bar{\alpha}]^{-1}$, $B = [\bar{\beta}]^{-1}$; peripheral loops use the stated whiskers	literal path development and conjugation-sensitive calibrations	textual path and base-generator conventions
D8	[1, §8.3 and Table 1]	M, N are oriented meridians based along y_1, s_2, with named correction paths	independent peripheral extractors; local normal-circle fibers; punctured-sweep checks	identification of target basing arcs with extracted arcs
D9	[1, §8.4]	reference T_α base direction is the y_1-based half-drift with word Ax	complement-only 72-letter identity certificate; side-sensitive controls	textual choice of reference half-drift
D10	[1, §9.1 and Table 1]	fixed member is $(m, n) = (0, 0)$ and uses displayed F1 and F2 slopes	independent relation-sheet extraction and exact comparison	textual member selection and relation-sheet transcription

continued on next page

Table S1 continued

	Target location	Audit interpretation	Mechanical or cited support	Residual assertion
D11	[1, Convention 2.1(C7), Remark 11.1]	ϵ_A, ϵ_B index algebraic relation sheets; fixed audit basis uses $(+,+)$	four finite certificates; convention-change lemma; sign-sensitive calibrations	textual status of signs and target-sheet identification
D12	[1, §8.7, Lemma 8.2]	target uses the two smooth Lagrangian tori represented by marked curve subbundles	local-flatness, normal-frontier, and topological-to-smooth transport certificates	smooth identification of target tori with audit tori
D13	[1, §§2.1,8.7,9]	geometric $+1$ surgeries use the F1/F2 Lagrangian-framing classes and, in the audit basis, kill $M\lambda_\alpha, N\lambda_\beta$	global split-form construction; Weinstein charts; relative Moser checks; framing-class independence	correspondence between target coefficient convention and fixed attaching maps
D14	[1, §§7.1–8.7]	marked bundle equivalence is an orientation-preserving diffeomorphism relative to the c, e collars and boundary marking	explicit PL homeomorphism and inverse; collar data; no smoothing theorem is claimed to discharge D14	smooth relative upgrade of the source-to-audit bundle equivalence

Proposition 1.1 (conditional sufficiency of the comparison checklist)**.** If every clause D1–D14 holds, then the marked smooth surgery data defining V_{aud} agree with the data defining Wuebben's fixed $(m, n) = (0, 0)$ member, and the filled manifolds are orientation-preservingly diffeomorphic.

Proof. Clauses D1–D11 identify the ordered marked fiber, monodromies, torus subbundles, based peripheral pairs, reference sheet, and filling slopes. D12–D14 supply the relative smooth bundle and torus identification and the agreement of Lagrangian framing classes. Conjugating the exterior boundary map by the two attaching maps carries meridian to meridian on each filling piece. The torus-filling extension lemma in the main paper, including its orientation calculation, therefore extends the comparison across both copies of $T^2 \times D^2$. This is only a sufficiency statement: it does not prove any clause D1–D14. □

2 Certificate inventory and replay

The normative manifest is `verification/luttinger/proof_certificates/manifest.json`, with SHA-256

`e8118b489a1b365c002a1931839fb419ab0f456aaecf724f2acf979612c9b5b9`.

It binds the sealed presentation `verification/luttinger/sealed_transport/r_presentations.json`, whose SHA-256 is

`1e5fc8cb6872a25fa45a227c1e1c207b288c861a3998fe3aae8722ea510c852c`.

The current working `verification/` directory has git tree object

`bc6806eb9b4065377d6194ff3797fbb02d58b386`.

The existence chain certificate `verification/luttinger/downstream_chain_certificate.json`, which binds `paper/main.tex` among its evidence files, has SHA-256

`49c1a7ad02a38cf26c7796a2cc4d471df3ed78c74da7c4c73348c20c0163fa29`.

The historical transfer chain, which is not a premise of any theorem, has SHA-256

`73257b9ea0883255bb15c70f77f551526eec1714a866ab9e645cfe374528e8ca`.

The manifest, sealed input, and verification-tree object are the operational root referenced by the finite theorem in the main paper. The two chain digests allow each separately labeled chain to be replayed without silently retaining an earlier manifest dependency.

2.1 The four algebraic sign specializations

For each of the four $n = 0$ entries defined in the finite theorem of the main paper, KBMAG produced a complete rewriting system in which all three numbered generators reduce to the identity. Completion is only the discovery method: each verdict is exported as a derivation DAG and proved by the Certificate Soundness Theorem in the main paper. Only the $(+, +)$ entry is geometrically identified there with V_{aud}; the other three are algebraic robustness checks. The certificate sizes are shown below.

system	signs	retained records
n=0	$(-,-)$	3620
n=0	$(-,+)$	3803
n=0	$(+,-)$	2470
n=0	$(+,+)$	6672

Retained derivation-DAG records over the sealed presentation. Each certificate verifies triviality from a fresh clone.

An earlier four-generator export, four adjacent $n = 1$ controls, calibration examples, deliberate corruptions, and the full framing-scan history are summarized in Section C. None carries logical load for the fixed $n = 0$ target.

2.2 Verifier architecture

Knuth–Bendix completion (KBMAG 1.5.9 [27], run under GAP 4.11.1 [28]) is used only as an *untrusted certificate generator*. A patched build of KBMAG records the ancestry of every equation it derives; the patch is distributed as `verification/luttinger/kbmag-proof/kbfns-proof.patch`. An untrusted compiler prunes that history to the dependency cone of the final identity rules and emits a certificate of the form in the Certificate Soundness Theorem in the main paper, binding the input presentation by SHA-256. The certified set is enforced, not assumed: both checkers reject a batch that verifies any case twice, and in full-inventory mode — the documented invocation — require the batch to be exactly the input's eight fillings, one file named by each case slug, over a source of the stated shape (three generators and 78 complement relators for the sealed chain, four and 95 for the earlier export) with two filling relators per case; the manifest generator refuses to hash a certificate directory that is not exactly one file per filling. A successful replay therefore verifies exactly the four n=0 sign pairs and the four n=1 controls. A small standalone checker then re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, tidying change, and final generator-to-identity rule; it neither imports nor runs KBMAG. A second checker, separately implemented in Ruby from the same specification, re-verifies all retained derivation records against the same hash-bound certificates and accepts, importing neither the first checker nor any project group code. All certificates, the exported rewriting systems, and a hash manifest replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KBMAG independently of the certificate path.

Reference verification algorithm. The main paper gives the normative four-record mathematical specification. For comparison with the implementations, its complete control flow is:

```
bind source bytes, SHA-256, case, relators, generator count, inverse table;
for each record in order:
  validate both words over the certificate alphabet;
  dispatch to inverse-axiom, input-relator, overlap, or change;
  replay every trace using only earlier record numbers;
  compare the required literal outputs and equation keys;
check one literal [letter] → [] root for every letter; accept.
```

The standalone file `verification/luttinger/CERTIFICATE_SPEC.md` fixes the concrete serialization and validation order.

The transport from the raw simplicial presentation to the presentation whose fillings are certified is sealed in the same sense. The canonical seeded run serializes the raw complement presentation (99,863 generators, 321,702 relators) together with its 89 tracked peripheral and coordinate words, records the 99,860 elementary eliminations that reduce it to the 3-generator, 78-relator presentation, and freezes that output. Two standalone checkers — one in Python, one in Ruby, importing neither the geometry nor the simplifier — replay the eliminations from those three frozen files alone: a step is accepted only if the eliminated generator occurs exactly once in the named current relator and the recorded replacement is the one that relator implies, input and output are bound by SHA-256, and the renumbered result must equal the frozen presentation relator by relator and tracked word by tracked word. Both reject corrupted-input, omitted-step, altered-substitution, and altered-output controls. The chain from the frozen raw complex to the eight triviality verdicts (39,163 retained derivation records over the sealed presentation) therefore replays end to end from committed files. The design principle throughout is that certificates are checked, not trusted: search and completion tools are untrusted generators, and every computational claim rests on a replayable artifact validated by small independent code.

The 78-relator input has $H_1 = \mathbb{Z}^2$. In the sealed generator order its relator exponent vectors span $\mathbb{Z}(0,0,1)$, while

$$[M] = [N] = 0, \qquad [\lambda_\alpha] = g_2, \qquad [\lambda_\beta] = -g_1.$$

Consequently the two-slope family $M\lambda_\alpha^p = N\lambda_\beta^q = 1$ has $H_1 \cong \mathbb{Z}/q \oplus \mathbb{Z}/p$ (with $\mathbb{Z}/0 = \mathbb{Z}$), as proved in the slope-family proposition of Section 5 of the main paper. Thus single fillings and infinitely many nonunit-coefficient fillings are visibly nontrivial. The four theorem-facing unit-coefficient fillings kill abelianization; their triviality rather than mere perfectness is carried by the derivation certificates. The negative controls and calibration examples in Section C show that the pipeline also produces nontrivial groups and that both checkers reject altered records and incomplete inventories. Those controls test implementation behavior; they are not substitutes for the soundness argument above.

This release adds the small checker `verification/luttinger/audit_manifold_invariants.py`. It binds the sealed input and replays the exponent-lattice, slope-family, Euler-characteristic, Betti-number, and mod-two fiber calculations promoted to the Audit-Manifold Theorem. The v2.3.0 manifest binds it. Its output distinguishes finite arithmetic from the topological hypotheses established in prose. This release also repairs Python's integer-letter check so that JSON booleans are rejected and expands its built-in corruption tests to match the Ruby checker's major failure classes. The external checker and geometry review guides are versioned with the repository but are not represented as completed independent reviews.

3 Geometric audit inventory

The geometric part of the main paper has a different trusted-computing boundary from the derivation certificates. The following table records the artifact statement moved out of the theorem stream. A successful replay proves the finite predicate in the third column; the last column names the mathematical conversion still performed in prose. Thus "checked" never silently means "formalized in a foundational proof assistant."

Table S2: Geometric artifact and trust inventory. Paths are relative to `verification/luttinger/`.

Layer	Artifact and checker	Finite replay claim	Remaining input
marked fiber	`independent_fiber_certificate.json`; `independent_fiber_audit.py` (473 lines)	closed genus-two surface, ordered five-chain, complementary faces, involution, and rotation systems pass exhaustive finite tests	oriented ribbon classification; periodic-disk theorem
relative bundle	`relative_marking_certificate.json` with its 249-line checker; `graph_clutching_certificate.json` with its 273-line checker	based monodromy, seams, inverse maps, marked sections, and both protected collars agree	mapping-cylinder gluing and relative isotopy
independent monodromy	`alternative_based_monodromy.json.gz`; `alternative_based_monodromy.py`	based path images and elementary moves replay in a separately subdivided stack	simplicial homotopy interpretation
local flatness	`torus_local_flatness_certificate.json`; `torus_local_flatness.py` (413 lines)	all 296 vertex, 888 edge, and 592 triangle link pairs have the recorded standard types	PL local-flatness criterion
normal frontier	`frontier_normal_equivalence_certificate.json`; `frontier_normal_equivalence.py` (312 lines)	all 113,336 mixed vertices, 769,256 comparabilities, and normal-circle fibers match	standard block-bundle meaning
complement	`complement_theorem_audit.py` (151 lines); sealed Tietze transport and its two checkers	full-subcomplex deformation data, cellular presentation, 99,860 eliminations, and 89 tracked words replay	cellular van Kampen and Tietze theory
peripheral data	`independent_peripheral_certificate.json`; `independent_peripheral_extractor.py` (939 lines); alpha and beta residual certificates with two checkers each	common whiskers, oriented dual circles, literal slopes, and both complement-only coordinate identities replay	geometric interpretation as meridians and product push-offs
drilled relation	`r3_complement_audit.py` (162 lines)	the full transport stack and basepoint homotopy avoid both tori	simplicial homotopy inside the complement
framing	`framing_check.py` (243 lines), `weinstein_chart_check.py` (257 lines), and `weinstein_chart_independence.py` (113 lines)	relative Moser, seam, exterior-algebra, momentum, Liouville-pairing, and first-jet identities hold	Lagrangian-neighborhood theorem and compact push-off isotopy
smooth bridge	`topological_smooth_bridge_certificate.json`; `topological_smooth_bridge.py` (181 lines)	orientation, collar, and compatibility hypotheses are present	D14 itself; no smoothing theorem is claimed to prove the relative smooth upgrade

The main paper suppresses literal edge identifiers and ancestry sizes so that its geometric argument is not interrupted by implementation inventory. For exact replay, the alpha meridian and longitude use the same four-edge whisker

$$y_1 = e_{24046} e_{90328} e_{79763} e_{72645}.$$

Put

$$w_\alpha = \texttt{lb\_a\_y1}^{-1}\texttt{geom\_Ageom\_x}, \qquad w_\beta = \texttt{lb\_b\_s2}^{-1}\texttt{geom\_r}^{-1}\texttt{geom\_Mgeom\_rgeom\_B}.$$

The complement-only target w_α has length 72 in the sealed input and is reduced to the identity by a 61-step target trace whose full ancestry cone contains 2,506 records. Its frozen source, compiler, certificate, and separate Python and Ruby verifiers are in `verification/luttinger/alpha_residual/`. The beta target w_β has length 113 and is reduced by an 82-step target trace with a 1,540-record ancestry cone; the analogous files are in `verification/luttinger/beta_residual/`. Each source contains the 78 sealed complement relators and no filling relation.

The line counts are descriptive rather than evidentiary. The release manifest binds exact bytes, and `proof_ledger.py` binds the logical dependencies. In particular, the finite predicates above do not resolve the source-comparison checklist D1–D14.

For compatibility with the frozen dependency certificate, its machine node retains the identifier `A_source_formalization_D` and the legacy phrase "Source Formalization D." In this revision both denote exactly the conjunction of checklist D1–D14; no frozen claim or digest is changed by the terminological refinement.

A The existence chain, and the historical transfer chain (attribution only)

The existence chain

The proof of Theorem A′ in the main paper is mirrored by a machine artifact, `verification/luttinger/downstream_chain.py`, which records every item of the argument as one of five kinds — an *external theorem* stated with its hypotheses and source; a *certificate* of this repository, bound by SHA-256; a written *proof* of the main paper, bound by SHA-256 of the source file; a *computed* fact, executed by the script; or a *step*, a deduction from earlier items — and freezes the result as `downstream_chain_certificate.json`. There is no assumption kind. The chain has 9 external theorems (van Kampen; duality and universal coefficients; the sublattice index formula; Wu's formula; Freedman; Luttinger surgery after Auroux–Donaldson–Katzarkov; asphericity of surface bundles; symplectic Kodaira dimension after Li and Liu; and Ho–Li), 4 certificates ($\pi_1(V_{\text{aud}}) = 1$; $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$; the framing identification; the seam identity of σ_{aud}), 3 bound proofs (the split form, the double, and the descent of the form), 5 computed facts, and 7 steps, ending in Theorem A′. The separately implemented Ruby checker `verify_downstream_chain.rb` re-derives every computed fact, checks that every citation resolves and the graph is acyclic, that the conclusion rests on the certified fundamental group, the certified involution, and the descent lemma, that every item lies on the conclusion's path, that no item is an assumption, that no item's text names Lidman–Piccirillo, Wuebben, or the clauses D1–D14, and that every bound file matches its digest. Both checkers reject five distributed negative controls.

The historical transfer chain

The deductions from $\pi_1(V) = 1$ to Theorems A, B, C occupy Part I of [1], whose Theorem 1.2 reduces them to the simple connectivity of V. The remainder of this appendix is historical and conditional: assume every clause D1–D14 above and thereby identify the audit-defined V_{aud} with this V. It carries the subsequent deductions through as an explicit dependency chain, now kept unchanged at `verification/luttinger/attribution/wuebben_transfer_chain.py` with its own certificate and Ruby checker; nothing on the existence path cites it. That chain is likewise a machine artifact: a script records every item as one of five kinds — an *external theorem* stated with its hypotheses and source; the explicit source-comparison *assumption* D; a *certificate* of this repository, bound by SHA-256; a *computed* fact, executed by the script; or a *step*, a deduction from earlier items — and freezes the result as a certificate. A second script, separately implemented in Ruby, re-derives every computed fact from scratch, checks that every citation resolves and that the dependency graph is acyclic, verifies that each of the three conclusions depends on the finite theorem of the main paper, and checks every evidence file against its recorded digest. The chain has 25 external theorems, one source-comparison assumption, 3 certificates, 15 computed facts, and 17 steps; its prose form is in the repository, and what follows is the mathematics, with the inputs first. Both scripts in this downstream pair were produced by Claude Fable 5; independence here likewise means separate implementations, not cross-model authorship.

External inputs

Elementary algebraic topology enters as: the Seifert–van Kampen theorem for a union along a connected bicollared subspace; the exact sequence $1 \to \pi_1(Z) \to \pi_1(W) \to G \to 1$ of a connected regular covering with deck group G; Poincaré–Lefschetz duality and universal coefficients, with the Euler characteristic as an alternating sum of Betti numbers over any field, multiplicative for bundles with compact fiber and additive along a common boundary; the index formula $\det(\mathrm{Gram}\, S) = [L : S]^2 \det(\mathrm{Gram}\, L)$ for a finite-index sublattice, with the unimodularity of the intersection form on H_2/Tors; Wu's formula $\langle w_2, x\rangle = x \cdot x$ and its consequence $\langle w_2(X), [F]\rangle = \chi(F) + F \cdot F \pmod 2$ for an embedded oriented surface; Novikov additivity of the signature; asphericity of bundles with aspherical fiber and base; the adjunction formula $K \cdot S + S \cdot S = 2g - 2$ for a symplectic surface; and the isotopy of any two orientation-preserving smooth embeddings of B^4 in a connected 4-manifold [26], which together with the amphichirality of the figure-eight knot makes smooth sliceness in a closed 4-manifold a diffeomorphism invariant.

The named theorems are the following.

- **Freedman** [11]: closed simply connected topological 4-manifolds are classified up to homeomorphism by the intersection form and the Kirby–Siebenmann invariant; every unimodular form is realized, uniquely if even (with $\mathrm{KS} \equiv \sigma/8$), and by exactly two manifolds if odd, one per value of KS.

- **Hambleton–Kreck** [12, 13, 14], in the form of [15, Theorem 5.1]: "Closed, oriented topological 4-manifolds with finite cyclic fundamental groups are classified up to homeomorphism by $\pi_1(M)$, q_M, the w_2-type, and $\mathrm{KS}(M)$," where q_M is the form on H_2/Tors and the w_2-type is (II) for spin M.

- **Thurston** [18] and the paper's Thurston-type form ([1], proofs of Proposition 1.3 and Lemma 8.2; [3], proof of Theorem 8): a closed surface bundle over a closed surface with homologically essential fiber is symplectic, and for $R \cup_\sigma R$ the form can be chosen positive on the fiber F and on the closed section $\widehat{\Gamma}$, with the surgery tori Lagrangian.

- **Luttinger surgery** [10, 4]: surgery on a Lagrangian torus yields a symplectic manifold, agreeing with the original outside the surgered neighborhood and with unchanged Euler characteristic.
- **Symplectic Kodaira dimension** [17], as summarized in [16, p. 1]: κ is defined from a minimal model and "is independent of the choice of symplectic form ω"; a minimal symplectic 4-manifold has $\kappa = -\infty$ iff it is rational or ruled, and rational and ruled 4-manifolds have $\pi_2 \neq 0$. **Ho–Li** [16, Theorem 1.1]: "The Luttinger surgery preserves the symplectic Kodaira dimension."
- **Symplectic Thom** [19]: an embedded symplectic surface in a closed symplectic 4-manifold is genus-minimizing in its homology class; and the construction ([1], proof of Lemma 4.3, after [22]) of a connected symplectic unbranched k-fold cover of a square-zero symplectic surface inside its tubular neighborhood, representing k times its class.
- **Klug** [23, Theorem 2]: "Let X^4 be a smooth compact connected oriented 4-manifold with ∂X an integer homology sphere. Let F^2 be an orientable characteristic surface with connected boundary that is properly embedded in X. Then $\mathrm{Arf}(F) + \mathrm{Arf}(\partial F) = (\sigma(X) - [F]^2)/8 + \mu(\partial X) \pmod 2$." For spin X, characteristic means vanishing in $H_2(X, \partial X; \mathbb{Z}/2)$. **Levine** [24]: $\mathrm{Arf}(K) = 0$ iff $\Delta_K(-1) \equiv \pm 1 \pmod 8$.
- **Slice disks and the 0-trace** ([3], proof of Theorem 1): a smooth slice disk with Seifert-framed normal bundle embeds the simply connected 0-trace, in which a capped Seifert surface is a closed square-zero surface generating H_2; the framing condition is automatic when the relative H_2 is torsion; the figure-eight knot has genus 1.
- **Lidman–Piccirillo's constructions** [3]: σ is a free orientation-reversing bundle involution (Lemma 4); the sections from the fixed points of φ_0 survive into V, their boundary circles are preserved by σ, and $R \cup_\sigma R$ is a closed genus-2 bundle over a closed genus-2 surface on which the surgeries act by Luttinger surgery on four disjoint tori missing F and $\widehat{\Gamma}$ (Lemma 6, proof of Theorem 8); $W = V/\sigma$ is a closed smooth oriented 4-manifold, double covered by Z, and is spin by the hyperbolic-pair argument of Lemma 7, whose hypotheses are the homology of V; the regluing diffeomorphism f of $S^3_0(Q)$ exists, and with $\mathcal{A} \subset S^3_0(Q)$ the framed annulus joining $\partial\Gamma$ to its image under $f \circ \sigma$, the closed surface $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma$ has odd square and meets F once (proof of Theorem 2); and the Floer-theoretic input, Lemmas 9 and 10 with the vanishing of all mixed invariants of $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ (and of its sum with any homology 4-sphere) along both square-zero lines, via splittings over $S^2 \times S^1$ (proof of Theorem 2, using [20, 21]).
- **Kawauchi's manifold** [25, 3]: B is a closed smooth oriented spin 4-manifold with $\pi_1(B) = H_1(B) = \mathbb{Z}/2$ and $b_2(B) = 0$, in which the figure-eight knot is smoothly slice.

The chain

Throughout, F is a fiber of R disjoint from the surgery tori, Γ a section through a fixed point of φ_0, $\widehat{\Gamma} = \Gamma \cup_\sigma \Gamma \subset Z$ and $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma \subset Z''$ its closures, $\mathcal{A}$ the framed annulus above. The finite calculations cited are those of the chain's certificate.

Proposition A.1. $H_1(V) = 0$, $H_3(V) = 0$, $H_2(V) = \mathbb{Z}\langle F\rangle$ with $F \cdot F = 0$, and V is spin. Moreover $\pi_1(Z) = \pi_1(Z'') = 1$.

Proof. Wuebben's Lemma 4.2 and the two filling relations give $H_1(V) = 0$ independently of simple connectivity; the certificate also implies it. The boundary $S^3_0(Q)$ is connected, so $H^1(V, \partial V) = 0$ and hence $H_3(V) = 0$, and $H_1(V, \partial V) = 0$ so $H_2(V)$ is torsion-free. $\chi(V) = \chi(R) = \chi(F)\chi(\text{base}) = (-2)(-1) = 2$, unchanged by the surgeries, and $b_0 = 1$, $b_1 = b_3 = b_4 = 0$ give $b_2(V) = 1$. A section is a properly embedded surface meeting F once, so $[F]$ is primitive and generates. $F \cdot F = 0$ since a fiber has trivial normal bundle. $H_2(V; \mathbb{Z}/2) = \mathbb{Z}/2\langle F\rangle$, w_2 is detected on it, and $\langle w_2, F\rangle = \chi(F) + F \cdot F = -2 + 0 \equiv 0$. Finally Z and Z'' are unions of two copies of V along the connected bicollared boundary, so by van Kampen each fundamental group is a quotient of $\pi_1(V) * \pi_1(V) = 1$. This is the one downstream step where simple connectivity of V, rather than only of Z, is needed: the regluing twists the amalgam. □

Proposition A.2. $H_2(Z) = \mathbb{Z}\langle F, \widehat{\Gamma}\rangle$ with intersection form the hyperbolic form H; Z is closed, simply connected, spin, with $b_2 = 2$ and signature 0; and Z is homeomorphic to $S^2 \times S^2$.

Proof. $\chi(Z) = 2\chi(V) - \chi(S^3_0(Q)) = 4$ and $\pi_1(Z) = 1$ give $b_2(Z) = 2$ with $H_2(Z)$ torsion-free. $F \cdot F = 0$, $F \cdot \widehat{\Gamma} = 1$ (a fiber meets a section once), and $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ by the certified self-intersection computation listed in the geometric inventory above. The Gram determinant is -1 and the form is unimodular, so the index formula makes $(F, \widehat{\Gamma})$ a basis and the form is H. H is even, so $w_2(Z) = 0$ by Wu's formula. For even forms $\mathrm{KS} \equiv \sigma/8 = 0$ is determined, so Freedman's classification gives $Z \cong S^2 \times S^2$. Note that spinness of Z is obtained here from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$; the paper's route through the spin quotient W is not needed for this step. □

Proposition A.3. Z is a closed symplectic 4-manifold in which F and $\widehat{\Gamma}$ are symplectic surfaces of genus 2 and square 0, and Z is not diffeomorphic to $S^2 \times S^2$.

Proof. $R \cup_\sigma R$ is a closed surface bundle whose fiber is homologically essential ($F \cdot \widehat{\Gamma} = 1$), so it carries the Thurston-type form, positive on F and $\widehat{\Gamma}$. The surgery tori are Lagrangian for it and the surgeries are the asserted Luttinger surgeries under the audit-model theorem together with checklist D1–D14, and Luttinger surgery preserves the form away from the surgered neighborhoods, which miss F and $\widehat{\Gamma}$. For the non-diffeomorphism: $R \cup_\sigma R$ is aspherical, hence minimal (an exceptional sphere would be null-homotopic, hence of square 0, not -1) and neither rational nor ruled (those have $\pi_2 \neq 0$), so $\kappa(R \cup_\sigma R) \neq -\infty$; Luttinger surgery preserves κ, so $\kappa(Z) \neq -\infty$. If $\psi\colon Z \to S^2 \times S^2$ were a diffeomorphism, composing with a reflection of one factor if necessary makes it orientation-preserving; ψ^* of a product form is then a symplectic form on Z inducing its orientation with $\kappa = \kappa(S^2 \times S^2) = -\infty$, contradicting the independence of κ from the form. □

Propositions A.2 and A.3 are Theorem A.

Proposition A.4. W is a closed oriented smooth spin 4-manifold with $\pi_1(W) = \mathbb{Z}/2$, $b_2(W) = 0$, signature 0, and $\mathrm{KS}(W) = 0$; and W is homeomorphic to B.

Proof. $Z \to W$ is a connected 2-fold covering, so $|\pi_1(W)| = 2$. $\chi(W) = \chi(Z)/2 = 2$ and $b_1(W) = 0$ give $b_2(W) = 0$, hence signature 0. More precisely, $H_1(W; \mathbb{Z}) = \mathbb{Z}/2$ and Poincaré duality give $H_3(W; \mathbb{Z}) = H^1(W; \mathbb{Z}) = 0$. The integral universal-coefficient sequence and duality then identify

$$H_2(W; \mathbb{Z}) \cong H^2(W; \mathbb{Z}) \cong \mathrm{Ext}(H_1(W; \mathbb{Z}), \mathbb{Z}) = \mathbb{Z}/2,$$

because $H_2(W;\mathbb{Z})$ has rank zero. Thus the vanishing of b_2 hides one torsion class that is used in the sliceness argument below. The manifold W is spin by Lidman–Piccirillo's Lemma 7, whose hypotheses Proposition A.1 supplies, and $\mathrm{KS}(W) = 0$ since W is smooth. W and B are then closed oriented topological 4-manifolds with the same Hambleton–Kreck invariants: $\pi_1 = \mathbb{Z}/2$, the zero form on $H_2/\mathrm{Tors} = 0$, w_2-type (II), $\mathrm{KS} = 0$. □

Proposition A.5. No nonzero square-zero class in $H_2(Z;\mathbb{Z})$ is represented by a smoothly embedded torus.

Proof. For $k \geq 1$ the connected symplectic k-fold covers of F and of $\widehat{\Gamma}$ represent kF and $k\widehat{\Gamma}$ and have genus $k+1$, since $\chi = -2k$; by the symplectic Thom theorem these genera are minimal, and reversing orientation covers $k < 0$. Since $(aF + b\widehat{\Gamma})^2 = 2ab$, every nonzero square-zero class is kF or $k\widehat{\Gamma}$ with $k \neq 0$, of minimal genus $|k| + 1 \geq 2$. □

Proposition A.6. The figure-eight knot is not smoothly slice in W.

Proof. Let D be a smooth slice disk in $W^\circ = W \setminus B^4$. The group $H_2(W^\circ, \partial; \mathbb{Z}) \cong H^2(W) \cong \mathbb{Z}/2$ is torsion, so the relative self-intersection of D is 0 and its framing is the Seifert framing. If $[D, \partial D] = 0$ in $H_2(W^\circ, \partial; \mathbb{Z}/2)$, then D is characteristic in the spin manifold W°, and Klug's theorem with $\mathrm{Arf}(D) = 0$, $\sigma(W^\circ) = 0$, $[D]^2 = 0$, $\mu(S^3) = 0$ forces $\mathrm{Arf}(4_1) = 0$; but $\Delta_{4_1}(t) = -t^2 + 3t - 1$ gives $\Delta_{4_1}(-1) = -5 \equiv 3 \pmod 8$, so $\mathrm{Arf}(4_1) = 1$.

It remains to make the other branch explicit. With $\mathbb{Z}/2$ coefficients, the long exact sequence of $(W^\circ, \partial W^\circ)$ and $H_2(S^3) = H_1(S^3) = 0$ gives

$$H_2(W^\circ; \mathbb{Z}/2) \xrightarrow{\cong} H_2(W^\circ, \partial W^\circ; \mathbb{Z}/2),$$

and Mayer–Vietoris for $W = W^\circ \cup_{S^3} B^4$ gives $H_2(W^\circ; \mathbb{Z}/2) \cong H_2(W; \mathbb{Z}/2)$. Consequently, if $[D, \partial D] \neq 0$, capping D by a Seifert surface in the removed ball produces a torus $T \subset W$ with $[T] \neq 0$ in $H_2(W; \mathbb{Z}/2)$. The Seifert framing gives $T \cdot T = 0$, equivalently the 0-trace $X_0(4_1)$ embeds in W.

The inclusion $i : X_0(4_1) \hookrightarrow W$ lifts through the connected cover $p : Z \to W$, because $\pi_1(X_0(4_1)) = 1$. For a chosen lift $\widetilde{i}$ one has $p \circ \widetilde{i} = i$; hence $\widetilde{i}$ remains an embedding and T lifts to a torus $\widetilde{T}$ on which p has degree one. Therefore

$$p_*[\widetilde{T}] = [T] \neq 0 \quad \text{in } H_2(W; \mathbb{Z}/2).$$

In particular $[\widetilde{T}] \neq 0$ in $H_2(Z; \mathbb{Z}/2) = H_2(Z; \mathbb{Z}) \otimes \mathbb{Z}/2$ (as $H_1(Z) = 0$), and a class with nonzero mod-2 reduction is nonzero in the torsion-free $H_2(Z; \mathbb{Z})$; and $\widetilde{T} \cdot \widetilde{T} = T \cdot T = 0$ because p is a local diffeomorphism. This contradicts Proposition A.5. □

Proposition A.4, Proposition A.6, the sliceness of the figure-eight knot in B, and the diffeomorphism invariance of sliceness give Theorem B.

Proposition A.7. *Z'' is closed, simply connected, smooth, with $b_2 = 2$, signature 0, and odd intersection form $\langle 1 \rangle \oplus \langle -1 \rangle$; hence Z'' is homeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

Proof. $\pi_1(Z'') = 1$ and $\chi(Z'') = 4$ give $b_2 = 2$ with H_2 free. Both Z and Z'' decompose into the same two oriented copies of V along their common boundary, so Novikov additivity gives $\sigma(Z'') = \sigma(Z) = 0$, independently of the regluing map. $F \cdot F = 0$, $F \cdot \Gamma'' = 1$, and $\Gamma'' \cdot \Gamma'' = 2\ell + 1$ is odd; the Gram determinant is -1, so (F, Γ'') is a basis, and $E = \Gamma'' - \ell F$, $D = F - E$ satisfy $E \cdot E = 1$, $D \cdot D = -1$, $E \cdot D = 0$ for every $\ell \in \mathbb{Z}$: the form is $\langle 1 \rangle \oplus \langle -1 \rangle$. Of Freedman's two manifolds with this form the one with $\mathrm{KS} = 0$ — Z'' is smooth — is $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. □

Proposition A.8. Z'' is not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

Proof. Z is symplectic with $|K_Z \cdot F| = 2$ by adjunction (F symplectic of genus 2 and square 0) and $H^2(V) = \mathbb{Z}$, so Lidman–Piccirillo's Lemma 9 gives $\Psi_{V,t} \neq 0$ for both spin^c structures with $|\langle c_1(t), F\rangle| = 2$. Z'' is $V \cup V$ along $S^3_0(Q)$ with t restricting to the non-torsion $s_\pm$, $b_2^+(Z'') = 1$, and span$\{F\}$ a square-zero line, so Lemma 10 gives a nonvanishing mixed invariant $\Phi_{Z'',\mathrm{span}\{F\},t}$. A diffeomorphism to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ would carry span$\{F\}$ to one of its two square-zero lines — in $\langle 1\rangle \oplus \langle -1\rangle$, $a^2 - b^2 = 0$ forces $a = \pm b$ — along each of which the manifold splits over $S^2 \times S^1$ and every mixed invariant vanishes. □

Propositions A.1, A.7 and A.8 are Theorem C. These deductions explain the original motivation; they are not theorems asserted by the main paper.

Three points where the chain's route differs from the paper's are worth recording. $H_1(V) = 0$ is taken from the certificate rather than from the paper's Lemma 4.2 and the surgery relations, which are needed only for the "forcing" direction of its Proposition 1.3. The intersection form of Z is identified from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$, so that Z is spin because its form is even; W spin still enters, for the Hambleton–Kreck invariants and for Klug's theorem. And the sliceness dichotomy is taken over $\mathbb{Z}/2$ throughout, which is exactly what makes the null-homologous case Klug's characteristic case and the lifted torus nonzero integrally.

B Trust boundary and data availability

Checker provenance and audit status

The verification was developed by Anthropic Claude models (Fable 5 and Opus 5) and OpenAI Codex (GPT-5 family) under the author's direction, with commit-level provenance and cross-model corrections retained in the public ledger. Mathematical responsibility rests with the human author; model provenance is not evidence of correctness.

Both filled-group checkers were produced as separate Python and Ruby implementations by OpenAI Codex. They share neither implementation code nor runtime libraries beyond their standard libraries, but they do share the published certificate specification and input format. Thus "independent" means implementation-independent, not independent human authorship or statistically uncorrelated errors. The downstream pair was separately implemented in Python and Ruby by Claude Fable 5. No theorem-critical checker has yet received an independent human line-by-line audit. The small source files, negative controls, and exact replay commands are public so that this remaining software-review task is explicit.

Inputs to the finite theorem and audit-model theorem

The finite theorem of the main paper has no mathematical input beyond its Certificate Soundness Theorem; its implementation trust rests on the two small checkers. The audit-model identification with the explicitly defined V_{aud} rests on standard inputs of PL topology: ribbon thickening and bistellar traces; classification of compact surfaces; local-flatness and normal-block criteria; and simplicial fundamental-group presentations with Tietze theory. The regular-neighborhood and local-flatness conventions are those of [6, Chapters 2–4, especially pp. 50–51] and [7, Chapters 3–5]; the periodic-disk input is [5, Theorem 3.1 and Proposition 3.2], and the torus-boundary linearization is [8, case (IV), pp. 464–465]. The presentation moves are the standard ones of [9]. The exact Lagrangian-neighborhood statement used for the Weinstein charts is the first paragraph of [4, §2.1]; the surgery and persistence of the symplectic form are [4, Definition 2.1 and Proposition 2.2]. No smoothing theorem is claimed to establish D14: that relative smooth comparison remains an explicit source hypothesis. This note applies rather than reproves the cited general results.

External specialist-review targets

No external PL or symplectic specialist audit is claimed. For the intrinsic simple-connectivity theorem, the highest-risk geometric steps are the marked bundle, normal-frontier, peripheral-word, and drilled-transport arguments in Sections 4–5 of the main paper. For the intrinsic symplectic conclusion, the highest-risk step is the Framing Bridge Theorem. The repository's external-review guide gives the former a claim-by-claim checklist. A review of the latter should answer the following six concrete questions against the named curves and boundary bases of the main paper:

(i) Does the global vertical area form descend through both clutching maps, does its sum with the base area form give the stated orientation, and can the two protected collars be standardized simultaneously without mixed terms?

(ii) Do the product push-offs of both named curves map to the exact Lagrangian-framing classes used by the stated surgery convention?

(iii) Do the relative Moser map, the alpha quotient, and the mapping-torus seam preserve the chosen orientations of the torus, momentum plane, and positive meridian?

(iv) After radial projection to the unit-covector boundary, does each constant-covector copy have zero coefficient in the meridian direction of the displayed integral boundary basis?

(v) Is one sufficiently small nonzero push-off defined over the entire named curve for the full fiber-dilation isotopy, and does the compact isotopy remain uniformly in the punctured neighborhood?

(vi) Do the smooth comparison assertions D12–D14 actually hold for the source object, so that these particular framing classes transport into the source surgery application? No smoothing theorem is claimed to answer this question automatically.

The main paper now states the boundary-class calculation, the compactness argument for the punctured isotopy, and the exact torus-filling extension matrix explicitly. These additions narrow the review target; they are not represented as an external audit.

Source comparison with Wuebben

Checklist D1–D14 is a separate evidence kind, not an external theorem and not a machine certificate. Its fourteen clauses split the marked-fiber, marked-bundle, peripheral/member, coefficient, and smooth-surgery comparisons into target-located assertions. Runs 51–56 and 68–69, the independent extractors, mutation controls, and the parsed author-script comparison give evidence for those readings; they cannot prove that a source diagram denotes the audit's marked smooth object. A future comparison $V_{\mathrm{aud}} \cong V$ and the historical downstream implications would depend on every clause; no theorem in the main article does.

Inputs to the historical conditional target application

The conditional application uses the twenty-five external theorems of Appendix A, each stated there with its hypotheses. The deep ones are Freedman's classification, the Hambleton–Kreck classification, the invariance theorems for symplectic Kodaira dimension (Li, Ho–Li), the symplectic Thom theorem, Klug's relative Rochlin theorem, and the Floer-theoretic lemmas of Lidman–Piccirillo with the Ozsváth–Szabó nonvanishing theorem behind them; the rest are elementary or are constructions of [3] that this note takes as stated. None is reproved. The repository's dependency ledger records the whole verification as an acyclic graph of 33 named external theorems, 37 machine certificates, 12 geometric arguments, one source-assumption node, and 21 derived claims ending in the three theorems, with every evidence file bound by hash. These are not competing counts: the 25/1/3/15/17 figures in Appendix A describe only the compact downstream chain, while the larger figures describe the project-wide ledger that also contains the model, source comparison, peripheral, framing, and PL verification.

Data availability

This verification targets Wuebben's arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

`16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a`

The target-version ledger in `TARGET.md` also records the Lidman–Piccirillo v1 digest. The immutable certificate set, frozen inputs, replay scripts, run transcripts, and provenance records are contained, with clean-checkout replay instructions, in the tagged release `v2.3.0` of the repository:

`https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v2.3.0`

The v2.3.0 verification snapshot is permanently archived at

`https://doi.org/10.5281/zenodo.22184725`

with concept DOI

`https://doi.org/10.5281/zenodo.22169753`,

which resolves to the newest archived version. The release deliberately does not redistribute three third-party source inputs: Wuebben's paper text, two files from his public code repository, and the Lidman–Piccirillo source figure. Replaying the corresponding source extractors requires supplying hash-matching copies as directed in `verification/IMPORT.md`; their frozen outputs and all subsequent comparisons ship. No certificate used for the finite or Audit-Manifold Theorem depends on downloading those files. The v2.3.0 manuscript and supplement correspond to that tagged release and its archival deposit. The working revision instead uses the manifest and verification-tree object printed at the start of this section; it is not an immutable release packet yet. The `verification/` tree of that release has git tree object hash

`d784d5502d414a206bd60e10c0ee995c26688e2c`

recoverable from a clean checkout as `git rev-parse v2.3.0:verification`; its filled-group proof manifest has SHA-256

`e8118b489a1b365c002a1931839fb419ab0f456aaecf724f2acf979612c9b5b9` and its downstream-chain certificate has SHA-256

`73257b9ea0883255bb15c70f77f551526eec1714a866ab9e645cfe374528e8ca`

The repository itself is public at

`https://github.com/VentiMath/exotic-s2xs2-verification`

The versioned release link above, rather than search-engine indexing, is the artifact to check. The raw completion histories from which certificates are compiled are reproducible build products and are omitted. The paper's own code is at `https://github.com/bwuebben/exotic-s2xs2`; nothing in the verification derives from it except where explicitly cited.

C Supporting computational history

The proof of the finite theorem of the main paper uses only the sealed three-generator chain. An earlier export of the same marked complement had four generators and 95 relators. Its four $n = 0$ certificates contain respectively 1123, 1075, 804, and 1504 retained records and reach the same verdicts, but the interpreter hash seed used to number its raw simplices was not recorded; its input Tietze trace therefore cannot be regenerated. It is retained as corroborating history and no ledger node depends on it.

Four adjacent $n = 1$ presentations replace Ax by $Ar^{-1} = Ax(rx)^{-1}$. They pass the same full-inventory certificate pipeline and check Wuebben's one-step re-indexing, but they are not used for the fixed $n = 0$ member. Wuebben's public sensitivity computations provide a particularly important negative control on interpretation: a 72-case nearby grid reports only trivial groups, and 14 of 15 single-word perturbations are silent at the group-decision layer [2]. Likewise, the framing-robustness scan here tests 96 cross-and-diagonal meridian shifts plus four zero-shift controls: 22 have derivation certificates and 78 have retained GAP session verdicts. All 100 were reported trivial. Together these experiments show that triviality is non-discriminating in this neighborhood. They do not validate any geometric dictionary or framing choice and are explicitly auxiliary.

The implementation has both positive and negative calibrations. On the standard T^4 Luttinger example it returns the complement $\mathbb{Z}^2 \times F_2$ and the textbook quotient $\mathbb{Z}^4$; on a Baldridge–Kirk example it separates the two sign conventions by trace fingerprints 1 and 3. A 96-relator common core completes to an infinite cyclic group, and a mapping-torus control distinguished $H_1 = \mathbb{Z}^2$ from the $H_1 = \mathbb{Z}^3$ identity case, exposing a construction error that was then fixed. Both filled-group checkers reject a wrong identity root, an altered input relator, an altered rewrite trace, a wrong generator count, an incomplete inventory, and a duplicated certificate batch. These tests show that the pipeline neither collapses every input nor accepts every file; the positive soundness claim remains the Certificate Soundness Theorem of the main paper.

References

[1] B. J. Wuebben, *An exotic $S^2 \times S^2$ and an exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$*, arXiv:2608.17267v1 (2026).

[2] B. J. Wuebben, *Public verification repository accompanying An exotic $S^2 \times S^2$ and an exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$*, commit `82fbf7ef4c2f75c75ff8e4239af8c258f244cdbc` (2026); see `scripts/pi1_grid.g`, `logs/pi1_grid.log`, `scripts/vsens.g`, and `logs/vsens_out.txt`, `https://github.com/bwuebben/exotic-s2xs2`.

[3] T. Lidman and L. Piccirillo, *Distinguishing closed 4-manifolds by slicing*, arXiv:2505.14387v1 (2025).

[4] D. Auroux, S. K. Donaldson, and L. Katzarkov, *Luttinger surgery along Lagrangian tori and non-isotopy for singular symplectic plane curves*, Math. Ann. **326** (2003), 185–203.

[5] A. Constantin and B. Kolev, *The theorem of Kerékjártó on periodic homeomorphisms of the disc and the sphere*, Enseign. Math. (2) **40** (1994), 193–204.

[6] C. P. Rourke and B. J. Sanderson, *Introduction to Piecewise-Linear Topology*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 69, Springer-Verlag, 1972.

[7] J. F. P. Hudson, *Piecewise Linear Topology*, W. A. Benjamin, 1969.

[8] A. E. Hatcher, *Linearization in 3-dimensional topology*, Proceedings of the International Congress of Mathematicians (Helsinki, 1978), pp. 463–468, Academia Scientiarum Fennica, 1980.

[9] R. C. Lyndon and P. E. Schupp, *Combinatorial Group Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 89, Springer-Verlag, 1977.

[10] K. M. Luttinger, *Lagrangian tori in* $\mathbb{R}^4$, J. Differential Geom. **42** (1995), 220–228.

[11] M. H. Freedman, *The topology of four-dimensional manifolds*, J. Differential Geom. **17** (1982), 357–453.

[12] I. Hambleton and M. Kreck, *On the classification of topological 4-manifolds with finite fundamental group*, Math. Ann. **280** (1988), 85–104.

[13] I. Hambleton and M. Kreck, *Cancellation, elliptic surfaces and the topology of certain four-manifolds*, J. Reine Angew. Math. **444** (1993), 79–100.

[14] I. Hambleton and M. Kreck, *On the classification of topological 4-manifolds with finite fundamental group: corrigendum*, Math. Ann. **372** (2018), 527–530.

[15] I. Hambleton, *Intersection forms, fundamental groups and 4-manifolds*, Proceedings of the 15th Gökova Geometry-Topology Conference (2008), 137–150.

[16] C.-I. Ho and T.-J. Li, *Luttinger surgery and Kodaira dimension*, Asian J. Math. **16** (2012), 299–318.

[17] T.-J. Li, *Symplectic 4-manifolds with Kodaira dimension zero*, J. Differential Geom. **74** (2006), 321–352.

[18] W. P. Thurston, *Some simple examples of symplectic manifolds*, Proc. Amer. Math. Soc. **55** (1976), 467–468.

[19] P. Ozsváth and Z. Szabó, *The symplectic Thom conjecture*, Ann. of Math. (2) **151** (2000), no. 1, 93–124.

[20] P. Ozsváth and Z. Szabó, *Holomorphic triangle invariants and the topology of symplectic four-manifolds*, Duke Math. J. **121** (2004), 1–34.

[21] P. Ozsváth and Z. Szabó, *Holomorphic triangles and invariants for smooth four-manifolds*, Adv. Math. **202** (2006), no. 2, 326–400.

[22] A. I. Stipsicz and Z. Szabó, *On the minimal genus problem in four-manifolds*, arXiv:2307.04202 (2023).

[23] M. R. Klug, *A relative version of Rochlin's theorem*, arXiv:2011.12418 (2021).

[24] J. Levine, *Polynomial invariants of knots of codimension two*, Ann. of Math. (2) **84** (1966), 537–554.

[25] A. Kawauchi, *Rational-slice knots via strongly negative-amphicheiral knots*, Commun. Math. Res. **25** (2009), 177–192.

[26] R. S. Palais, *Extending diffeomorphisms*, Proc. Amer. Math. Soc. **11** (1960), 274–277.

[27] D. F. Holt, *KBMAG — Knuth–Bendix in Monoids and Automatic Groups*, software package, version 1.5.9.

[28] The GAP Group, *GAP — Groups, Algorithms, and Programming*, `https://www.gap-system.org`.